

Muhammad Talha Zafar

Mechanical Engineering Student | Embedded Systems & Robotics | Digital Marketing

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PROFILE

Mechanical Engineering student at NUST Islamabad with hands-on experience in embedded systems, robotics, and digital marketing. Currently interning at NASTP Rawalpindi in the Design Department, gaining practical exposure to professional engineering workflows. Independently built and executed a B2B digital marketing strategy for an industrial robotics brand, managing CRM automation, email campaigns, and competitive SEO analysis. Combines strong technical communication with a data-driven marketing approach — well-suited for a Digital Marketing role in a medical or healthcare institution seeking to grow its online presence and strengthen its brand. Achieved A+ in both Matriculation and Intermediate.

EDUCATION

Bachelor of Engineering — Mechanical Engineering

2025 – Present

National University of Science and Technology (NUST), Islamabad

Pakistan's #1 ranked engineering university. Focus areas: precision design, applied mechanics, and engineering drawing.

Higher Secondary Certificate (Pre-Engineering)

2023 – 2025

Govt. Comprehensive Higher Secondary School, Sahiwal • Grade: **A+**

Strong foundation in Mathematics, Physics, and Chemistry.

Secondary School Certificate (Science)

2020 – 2022

Sanabil Educational Complex, Dad Fatyana, Chichawatni • Grade: **A+**

Grounding in Physics, Chemistry, Mathematics, and Biology.

WORK EXPERIENCE

Design Department Intern | NASTP, Rawalpindi

2026

- Gaining hands-on exposure to professional engineering design workflows within one of Pakistan's leading aerospace and technology parks.
- Assisting the design team with technical drawings, CAD modelling, and design documentation using industry-standard tools.
- Collaborating with senior engineers on design reviews, contributing to iterative prototyping and quality checks.

Digital Marketing & Automation Specialist | MZ Robotics — Freelance / Self-Employed

2026

- Engineered a full B2B digital marketing funnel for an industrial robotics brand (MZ Robotics), covering brand identity, lead generation, and customer acquisition strategy.
- Built and managed an automated email nurturing campaign in HubSpot CRM; designed 5-step drip sequences targeting technical buyers, achieving an estimated 30%+ open rate.
- Conducted zero-budget competitive analysis of the industrial robotics sector using SimilarWeb; identified 5 high-value keyword gaps and authored a Strategic Positioning Report that informed organic SEO strategy.
- Bridged the gap between complex engineering specifications and customer-facing content, producing technical marketing copy that drove qualified lead engagement.

PROJECTS

Autonomous Pipe-Climbing Robot *Robotics / Mechanical Design / ESP32*

2025

- Engineered a vertical climbing robot using an ESP32-based control system and high-torque motors; developed a spring-loaded 4-wheel mechanism to maintain constant grip force against varying pipe diameters.

Automatic Parking System *Arduino / Embedded Systems / C++*

2024

- Designed and prototyped an automated parking solution using Arduino Uno, IR sensors, servo motors for gate control, and I2C LCD for real-time slot availability — reducing manual overhead for up to 4 simultaneous bays.

IoT-Based Automated Irrigation System *Arduino / IoT / C++*

2024

- Engineered a self-regulating watering system using soil moisture sensors and a C++ logic controller that triggers a 5V relay and DC pump based on real-time soil data, reducing water waste.

Driver Drowsiness Alarm *Embedded Systems / C++ / Wearable Tech*

2024

- Prototyped wearable glasses with IR sensors to detect eye closure; designed a C++ logic loop to trigger an audible alarm within 200ms of microsleep detection for real-time road safety enhancement.

Autonomous Line-Following Robot (LFR) *Robotics / Arduino / Embedded Systems*

2024

- Engineered a high-speed autonomous robot using an Arduino Uno and IR sensor array; tuned PID-style control logic to maintain precision path tracking at variable speeds.

Rube Goldberg Engineering Challenge *Mechanical Design / Prototyping*

2024

- Designed a multi-stage Rube Goldberg machine demonstrating energy transfer, mechanical advantage, and structural stability through iterative prototyping and physics-based sequencing.

SKILLS & TOOLS

CAD & Engineering: AutoCAD, Fusion 360, SolidWorks, Proteus, Engineering Drawing, 3D Modeling

Embedded & Robotics: Arduino IDE, Arduino Uno, ESP32, PlatformIO, ROS 2, IR Sensors, Servo Motors, Relay Control

Programming: C / C++, Python, HTML / CSS, Java

Digital Marketing & CRM: HubSpot CRM, Mailchimp, SEO & Competitive Analysis, B2B Marketing, Email Automation

Software: VS Code, Dev C++, Microsoft Office (Word, Excel, PowerPoint)

Soft Skills: Technical Communication, Problem-Solving, Iterative Prototyping, Cross-functional Reporting

AWARDS & ACHIEVEMENTS

1st Place — District STEM Competition (Robotics & Engineering Category) *District Sahiwal, 2025*

- Won district-level STEM competition judged on prototype functionality, engineering design, and innovation; competed against teams from across Sahiwal district.